

(Very Short) Introduction to Propositional Logic and Basic Logical Operators

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Logic is important for:

- Basic Programming
 - Digital circuits / logic gates in electronics
 - 0 and 1 are basically True and False!
 - Knowing when people are bullshitting online
 - So we don't get tricked!
 - So much more!
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- **Examining arguments in Philosophy**

Logic & Arguments

Premise 1: All dogs have legs

Premise 2: Bella is a dog

Conclusion: Thus, Bella has legs

Premise 1: All dogs have legs

Premise 2: I have legs

Conclusion: Thus, I must be a dog



Image source:
<https://www.rover.com/blog/dog-names/>



Validity & Soundness

- DEF: We say an argument is logically **valid** when its conclusion has to follow its premises
 - Valid argument: If premises , then conclusion 
 - Invalid argument: Possible that premises  and conclusion 
- DEF: We say an argument is **sound** when it's valid and all of its premises are true.

Coming back

Premise 1: All dogs have legs

Premise 2: Bella is a dog

Conclusion: Bella has legs

Valid?

Sound?

Premise 1: All dogs have legs

Premise 2: I have legs

Conclusion: I must be a dog

Valid?

Sound?

Coming back

Premise 1: All dogs have legs

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Conclusion: Bella has legs 

Valid? 

Sound? 

Premise 1: All dogs have legs

Premise 2: I have legs

Conclusion: I must be a dog 

Valid? 

Sound? 

Only when an argument is sound can we guarantee that its conclusion is true!

Propositional Logic

DEF: A proposition is (loosely) a statement that is true or false, but not both.

Examples:

- The earth orbits the sun  TRUE Proposition
- The sun is a planet  FALSE Proposition
- There are aliens outside of the solar system  Proposition which we don't know true or not
- But aliens can't be both there and not there

Propositional Logic : a variant of Symbolic Logic

P: The earth orbits the sun (But basically any symbol you want!)

Q: The sun is a planet

C: Earth orbits a planet

- If the earth orbits the sun and the sun is a planet, then earth orbits a planet.
- $P \wedge Q \rightarrow C$ (We'll know how to interpret after some lectures!)

Question: Is this argument logical valid? Is it sound?

Propositional Logic : a variant of Symbolic Logic

P: The earth orbits the sun (But basically any symbol you want!)

Q: The sun is a planet

C: Earth orbits a planet

- If the earth orbits the sun and the sun is a planet, then earth orbits a planet.
- $P \wedge Q \rightarrow C$ (You'll know how to interpret after some lectures!)

Answer: This argument is valid but not sound (Q is false)

Logical Operator: Not $\neg$ (Negation)

P	$\neg P$
T	F
F	T

Example:

P: I will get an A in this class

$\neg P$: I will not get an A in this class

Logical Operator: Or $\vee$ (Disjunction)

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

$P \vee Q$ is true as long as one of P or Q is true

Example: $\neg P \vee Q$ (Not P or Q)

$\neg$ has the highest priority!

P	Q	$\neg P$	$\neg P \vee Q$
<u>F</u>	T	F	T
T	F	F	F
<u>F</u>	T	T	T
F	F	T	T

Logical Or and exclusive Or

- If a student scores above 90% in the final or has 90% in all homeworks, they pass the class
- What happens to a student who both scores 90% in final and has 90% in all homeworks?
- James is now either in the library or at the sports field
- Can James be both at the library and at the sports field?

Exclusive Or: true when one but not both of A and B is true.

Bullet Points

- Valid and Sound arguments
- Propositions
- Not $\neg$ (Negation)
- Or $\vee$ (Disjunction)
- Draw truth tables

Practice Questions:

1. Complete the following truth table:

Argument is valid	All premises of the argument are true	Argument is sound
T	T	
T	F	
F	T	
F	F	

Practice Questions:

2. Draw the truth table for $\neg(P \vee Q)$. You can read this as “It’s not the case that P or Q”.

3. Draw the truth table for exclusive or

P	Q	Exclusive Or
T	T	
T	F	
F	T	
F	F	

Challenge Question:

- In the next class we are going to talk about $\wedge$ “and”, which has truth table below. We say that $\neg$ “not” and $\vee$ “or” are logically complete, meaning any logical operator can be expressed with only these two. How would you express “and” with $\neg$ and $\vee$? Demonstrate by showing that your expression has the same truth values as “and”.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Practice Questions 1 Answer

1. Complete the following truth table:

Argument is valid	All premises of the argument are true	Argument is sound
T	T	T
T	F	F
F	T	F
F	F	F

Practice Question 2 Answer

P	Q	$P \vee Q$	$\neg (P \vee Q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

Practice Question 3 Answer

P	Q	Exclusive Or
T	T	F
T	F	T
F	T	T
F	F	F

Challenge Question Answer

$$\neg(\neg P \vee \neg Q)$$

P	Q	$\neg P$	$\neg Q$	$(\neg P \vee \neg Q)$	$\neg(\neg P \vee \neg Q)$
T	T	F	F	F	T
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	F